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Review article

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Mild autonomous cortisol secretion: pathophysiology, comorbidities and management approaches

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Abstract

The majority of incidentally discovered adrenal tumours are benign adrenocortical adenomas and the prevalence of adrenocortical adenomas is around 1–7% on cross-sectional abdominal imaging. These can be non-functioning adrenal tumours or they can be associated with autonomous cortisol secretion on a spectrum that ranges from rare clinically overt adrenal Cushing syndrome to the much more prevalent mild autonomous cortisol secretion (MACS) without signs of Cushing syndrome. MACS is diagnosed (based on an abnormal overnight dexamethasone suppression test) in 20–50% of patients with adrenal adenomas. MACS is associated with cardiovascular morbidity, frailty, fragility fractures, decreased quality of life and increased mortality. Management of MACS should be individualized based on patient characteristics and includes adrenalectomy or conservative follow-up with treatment of associated comorbidities. Identifying patients with MACS who are most likely to benefit from adrenalectomy is challenging, as adrenalectomy results in improvement of cardiovascular morbidity in some, but not all, patients with MACS. Of note, diagnosis and management of patients with bilateral MACS is especially challenging. Current gaps in MACS clinical practice include a lack of specific biomarkers diagnostic of MACS-related health outcomes and a paucity of clinical trials demonstrating the efficacy of adrenalectomy on comorbidities associated with MACS. In addition, little evidence exists to demonstrate the efficacy and safety of long-term medical therapy in patients with MACS.

Sections

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Key points

- Mild autonomous cortisol secretion (MACS) is diagnosed based on the 1 mg overnight dexamethasone test and is found in 20–50% of patients with adrenal adenomas lacking signs and symptoms of Cushing syndrome.
- Patients with adrenal adenomas show distinct changes in the steroid and global metabolome, which correlate with the degree of cortisol excess across MACS and Cushing syndrome.
- MACS is associated with an increased likelihood of having cardiovascular risk factors and an increased risk of mortality.
- Management of MACS must be individualized based on patient characteristics; the options range from adrenalectomy to long-term follow-up and conservative management of comorbidities.
- Post-operative adrenal insufficiency is seen in around 50% of patients with MACS who undergo unilateral adrenalectomy.

Introduction

An adrenal incidentaloma is an adrenal mass detected on imaging that is performed for reasons unrelated to suspected adrenal disease[1]. In individuals with adrenal incidentalomas, mild autonomous cortisol secretion (MACS) is the most common hormonal abnormality observed, the prevalence of which far exceeds that of adrenal Cushing syndrome[1,2]. MACS is diagnosed based on an abnormal serum level of cortisol >50 nmol/l (>1.8 μg/dl) following the 1 mg overnight dexamethasone suppression test (1 mg-DST)[1]. This condition is associated with a high cardiometabolic burden, increased risk of cardiovascular events and increased mortality[3–5]. Data published in the past decade show that patients with benign non-functioning adrenal tumours (NFAT) are also at higher cardiovascular risk than patients without adrenal adenomas, suggesting that even subtle autonomous cortisol secretion is clinically important[4,6,7].

In this Review, we provide a brief overview of the aetiology and epidemiology of adrenal incidentaloma and the characteristics and potential mechanisms of MACS. Furthermore, we synthesize the emerging data on metabolomics in patients with MACS. We discuss the diagnosis, associated comorbidities and management of this common condition, with a special focus on bilateral MACS. Finally, we provide guidance for post-operative management of affected individuals.

Adrenal incidentaloma: aetiology and epidemiology

Adrenal incidentalomas are fairly common, with 1.4–7.0% of adults and 10% of those older than 70 years diagnosed with adrenal incidentalomas during abdominal CT and MRI studies[8,9]. The incidental discovery of small and benign adrenal masses in older adults during cross-sectional imaging has driven the 10-fold increase of adrenal tumour incidence over the past two decades, whereas the incidence of large and malignant adrenal tumours has not changed[2].

In Western populations, adrenal incidentalomas are more common in women than in men (58–64% of cases are in women) with an average age at the time of adrenal mass diagnosis of 58–61 years[2,3,5,10,11]. By contrast, in Asian populations, men (51–57%) and younger individuals (55–58 years) are more frequently affected[9,12,13].

The distribution of aetiologies among adrenal incidentalomas varies considerably based on the setting (for example, endocrinology specialist centres, population-based studies or unselected populations) (Supplementary Fig. 1). However, benign adrenocortical adenomas represent 80–96% of adrenal incidentalomas regardless of the population studied[1,2,9,10]. The prevalence of hormone excess in adrenal incidentalomas depends on the biochemical definitions used in the literature. Overall, MACS is the most common hormonal abnormality detected in individuals with adrenal incidentalomas (19–50%), whereas 40–70% have NFAT, 2–12% have primary aldosteronism and 1–4% have Cushing syndrome[1,9] (Supplementary Fig. 1).

Several large-scale studies have shown that compared with NFAT, MACS is more frequently diagnosed in women (66–69% for MACS versus 63–64% for NFAT), and affected individuals are older (median age 63–64 years for MACS versus 58–60 years for NFAT) and more frequently present with larger (median diameter 27–33 mm MACS versus 22–25 mm NFAT) and bilateral tumours (22–30% MACS versus 12–17% NFAT are bilateral)[3–5]. However, a cross-sectional study in a large Chinese cohort undergoing adrenal CT found a higher prevalence of MACS in men (57% with MACS were men) and affected individuals had considerably smaller tumours (median diameter 16.5 mm), which were bilateral in only 12.5% of individuals[9].

Characteristics of MACS

Adrenal adenoma or primary bilateral macronodular adrenal hyperplasia?

Benign adrenocortical nodular disease associated with MACS can present more frequently as adrenocortical adenomas (either unilateral or bilateral) or more rarely as primary bilateral macronodular adrenal hyperplasia (PBMAH). Adrenocortical adenomas are typically homogeneous, well-circumscribed, lipid-rich nodules of the zona fasciculata with benign histological characteristics[14]. Atrophy of the adjacent cortex can be observed and this effect is associated with the lack of pituitary-secreted adrenocorticotropic hormone (ACTH) adrenal stimulation, which occurs owing to negative feedback from the adrenal-derived autonomous cortisol secretion[15,16]. By contrast, PBMAH is characterized by bilateral enlargement of the adrenal glands, with multiple cortical nodules >1 cm containing smaller cells with lower lipid content and larger fasciculata-like cells with higher lipid content than normal adrenal tissue. Diffuse internodular hyperplasia is typically observed in PBMAH, both histologically and radiologically[17].

Distinguishing bilateral adenomas from PBMAH associated with MACS can be challenging, as many patients with MACS do not undergo surgery, hence histology is lacking, and genetic testing for germline mutations associated with PBMAH is not carried out routinely. Considering that patients with MACS more frequently have bilateral adrenal masses than those with NFAT[3–5], some patients with MACS might potentially have undiagnosed PBMAH and this condition might be more frequent than previously thought[17].

Tumour growth and biochemistry

The mechanisms underpinning the development of MACS in most patients, both in terms of tumour growth and hormone hypersecretion, remain elusive (Fig. 1). MACS and adrenocortical adenomas causing overt Cushing syndrome share similar biochemistry, that is, ACTH-independent autonomous cortisol secretion; however, the epidemiology and biology of MACS are closer to NFAT than adrenal Cushing syndrome. Similar to NFAT, MACS mostly affects women of post-menopausal age, whereas adrenocortical adenomas causing Cushing syndrome tend to be diagnosed in younger women[4,18]. Apart from incidentally discovered PBMAH, which can progress to overt Cushing syndrome if sufficiently large adrenal nodules develop over time[17], the risk of progression from MACS to overt Cushing syndrome is vanishingly low[1,3]. By contrast, 4–12% of patients with NFAT can develop an abnormal 1 mg-DST during follow-up[3,19–21] (Fig. 2). Furthermore, the cardiometabolic burden across NFAT and MACS exists on a spectrum, with higher risk with increasing serum cortisol values during the 1 mg-DST, even below the 'normal' cut-off of >50 nmol/l (>1.8 μg/dl)[4,7,22–26]. Multisteroid profiling showed higher glucocorticoid and lower androgen adrenal output in patients with NFAT compared with those without adenomas (Supplementary Fig. 2). This finding suggests that at least a subset of patients with NFAT could have a subtle degree of autonomous cortisol secretion that is not identified by the current diagnostic test for cortisol excess, the 1 mg-DST.

Genetic factors

In line with these observations, unilateral sporadic NFAT and MACS adenomas share similar genomic and transcriptomic signatures, which clearly separate them from adrenal Cushing syndrome[27–30]. Somatic *CTNNB1* mutations leading to constitutive Wnt–β-catenin pathway

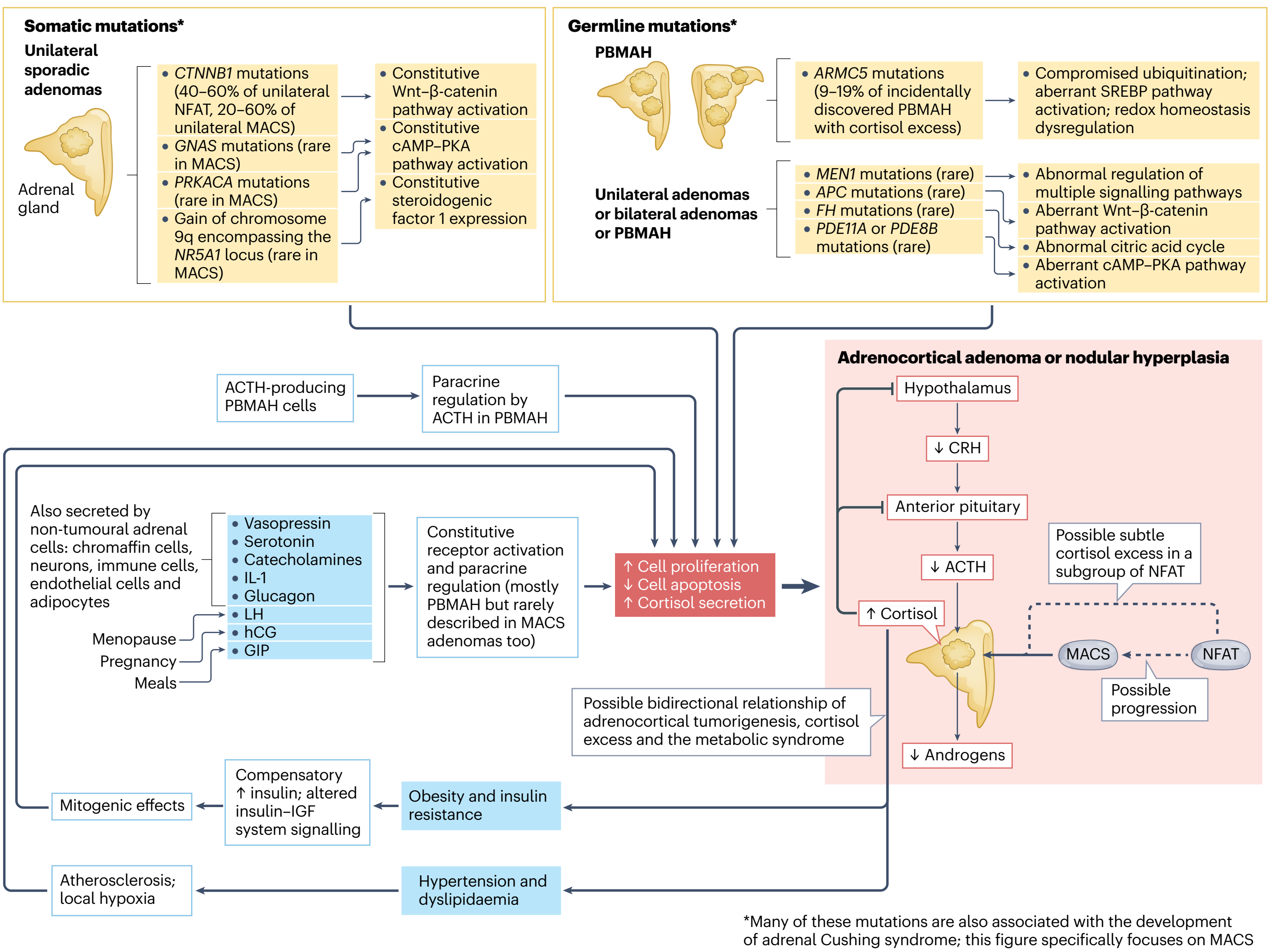

Fig. 1 | Pathophysiology of mild autonomous cortisol secretion. Adrenocortical adenomas or nodular hyperplasia associated with mild autonomous cortisol secretion (MACS) is characterized by chronic and low-degree increased glucocorticoid output, which results in reduced secretion of hypothalamic corticotropin releasing hormone (CRH), pituitary adrenocorticotropic hormone (ACTH) and adrenal androgens. Unilateral sporadic non-functioning adrenal tumour (NFAT) and MACS share a similar genetic landscape, with somatic *CTNNB1* mutations being a common finding. Several germline mutations are also associated with unilateral adenomas, bilateral adenomas or primary bilateral macronodular adrenal hyperplasia (PBMAH), with or without cortisol excess (both MACS and Cushing syndrome). *ARMC5* pathogenic variants are the most common germline mutations observed in incidentally discovered PBMAH with cortisol excess. The figure does not include mutations only associated with micronodular adrenal hyperplasia and cortisol excess (that is, no visible adrenal masses), mutations only observed in aldosterone-producing adenomas or adrenal Cushing syndrome, *NR3C1* pathogenic variants (partial glucocorticoid resistance associated with bilateral adrenal nodular hyperplasia) and adrenocortical hyperplasia associated with ACTH-dependent Cushing syndrome. Other factors potentially contributing to the development of adrenocortical adenomas or hyperplasia and increased cortisol secretion are also shown. GIP, glucose-dependent insulinotropic polypeptide; hCG, human chorionic gonadotropin; IGF, insulin-like growth factor; LH, luteinizing hormone; PKA, protein kinase A; SREBP, sterol regulatory-element binding protein.

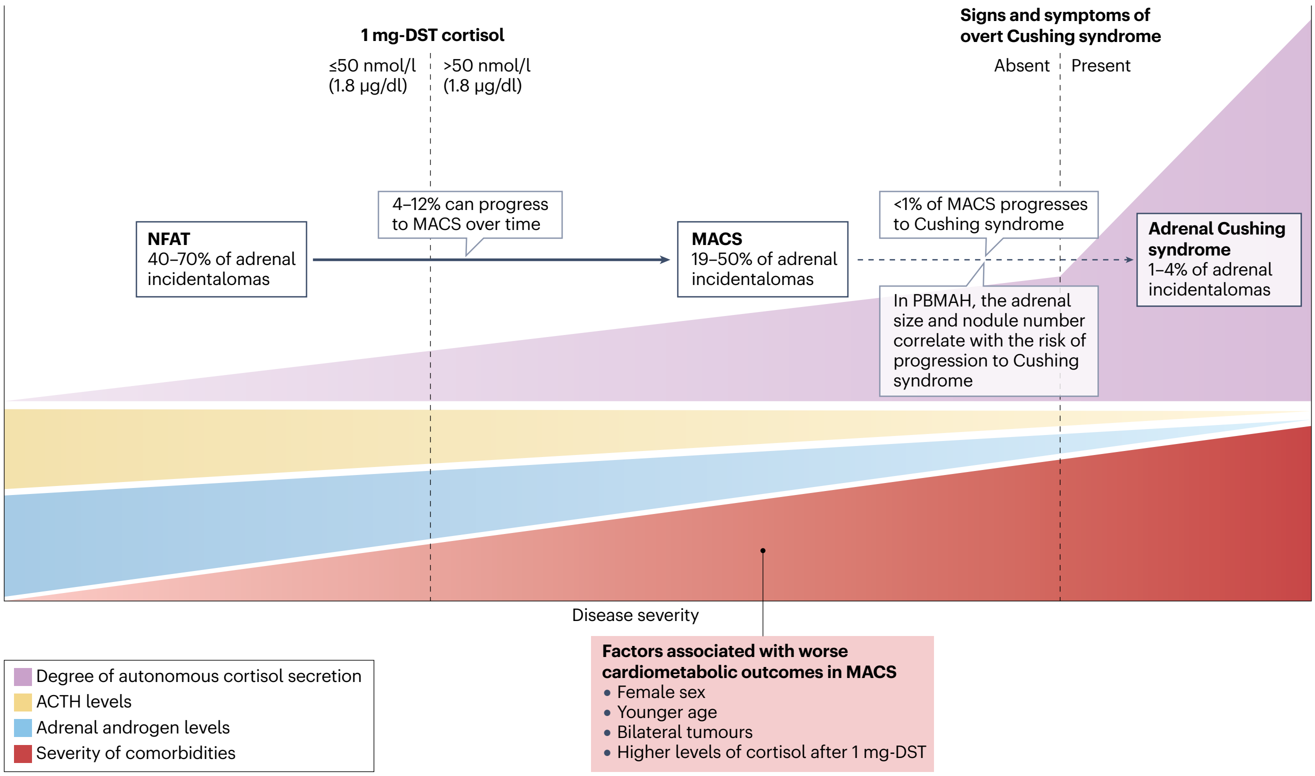

Fig. 2 | Natural history of mild autonomous cortisol secretion. Both mild autonomous cortisol secretion (MACS) and Cushing syndrome are characterized by adrenocorticotropic hormone (ACTH)-independent autonomous cortisol secretion, with MACS presenting with lower degrees of autonomous cortisol secretion and lacking the distinct clinical features of full-blown Cushing syndrome. Some patients with non-functioning adrenal tumour (NFAT) can also have a subtle degree of autonomous cortisol secretion and can progress to MACS over time. Progression from MACS to Cushing syndrome is exceptionally rare, except in the context of germline mutations causing primary bilateral macronodular adrenal hyperplasia (PBMAH), where the gradual increase in the size and number of adrenal nodules result in higher autonomous cortisol secretion. 1 mg-DST, 1 mg overnight dexamethasone suppression test.

activation are by far the most common genetic abnormality found in unilateral NFAT and MACS adenomas[31–33] (Fig. 1). By contrast, autonomous cortisol secretion in the majority of unilateral adrenocortical adenomas causing Cushing syndrome is driven by somatic mutations that lead to constitutive cAMP–PKA pathway activation[32,34,35]. Wnt–β-catenin signalling is critical for the development and maintenance of the adrenal cortex, and its role in adrenal tumorigenesis is well established and related to increased cell proliferation and reduced apoptosis[36].

Benign adrenocortical nodular disease associated with MACS is also observed in several germline mutations[8]. For example, pathogenic variants of the tumour suppressor gene *ARMC5* are observed in 9–19% of individuals with incidentally discovered PBMAH and MACS[17,37,38]; therefore, genetic testing for germline pathogenic *ARMC5* variants should be considered in the presence of multiple adrenal nodules and ACTH-independent MACS[1,39]. Unilateral or bilateral benign adrenocortical nodular disease with and without MACS is also observed in multiple endocrine neoplasia type 1 (*MEN1* mutations), familial adenomatous polyposis (*APC* mutations) and hereditary leiomyomatosis and renal cell cancer syndrome (*FH* mutations)[8] (Fig. 1).

Other potential mechanisms

Other factors have been implicated in the development of adrenocortical nodules and MACS, but their role remains mostly speculative (Fig. 1). In PBMAH, the aberrant expression of various G protein-coupled receptors regulating cell proliferation and steroidogenesis is well documented[17,40]. In some unilateral MACS adenomas, evidence also exists of increased cortisol secretion in response to stimuli including vasopressin[41,42], gastric inhibitory polypeptide[43], upright posture[41], serotonin[41,44], glucagon[41,44], angiotensin II[41] and gonadotropin-releasing hormone[41,44,45]. Increased cortisol secretion in response to gonadotropin-releasing hormone analogues suggests aberrant expression of the luteinizing hormone–choriogonadotropin receptor by the tumour. This suggestion is particularly interesting considering that most women with MACS are post-menopausal and have high circulating levels of luteinizing hormone, which might thus contribute to adrenal tumorigenesis and increased cortisol secretion[46–50]. Similarly, paracrine regulation modulates cell proliferation and steroid secretion in the adrenal cortex; therefore, non-tumoural adrenal cell populations could potentially have a role in the physiopathology of MACS[51–53].

Another potential mechanism to consider is the crosstalk between manifestations of the metabolic syndrome and adrenal tumorigenesis. Insulin stimulates the proliferation of adrenocortical cells[54] and in patients with NFAT or MACS, the degree of insulin resistance was positively correlated with tumour diameter[49,55]. Furthermore, changes in vascular supply to the adrenal cortex (owing to atherosclerosis or hypertension) have been hypothesized to cause local hypoxia and compensatory cell proliferation[56]. It is therefore intriguing to speculate a direct contribution of insulin resistance, hyperinsulinaemia and hypertension to adrenocortical adenoma and/or hyperplasia development, and the resulting vicious cycle of subtle cortisol excess that further fuels the manifestations of the metabolic syndrome[57–59] (Fig. 1).

Metabolic perturbations

Several metabolic changes have been observed in patients with adrenocortical tumours and different degrees of autonomous cortisol secretion. Most studies have focused on systemic steroid metabolism perturbations, but data are emerging on the effect of cortisol excess on intratumoural metabolism and other metabolic processes.

Multisteroid profiling

Steroid biosynthesis and metabolism are reflected by the serum steroid metabolome. Even more detail can be provided by the 24-h urine steroid metabolome, which can impart valuable insights into alterations of steroid flow and output (Fig. 3).

Analysis of the urine steroid metabolome of NFAT, MACS and adrenal Cushing syndrome shows changes that are consistent with the increasing degrees of ACTH-independent autonomous cortisol secretion[4,60–71] (Supplementary Fig. 2a). Serum and plasma steroid metabolome profiling of these patient populations also show varying degrees of glucocorticoid excess and androgen suppression[70,72–84] (Supplementary Fig. 2b). For example, compared with individuals with no steroid excess or NFAT, those with MACS or Cushing syndrome show a progressive decrease in the levels of androgens and their metabolites. Conversely, a progressive increase is observed in the levels of glucocorticoid metabolites, which is most pronounced in patients with Cushing syndrome, as well as of the immediate cortisol precursor 11-deoxycortisol and its metabolite tetrahydro-11-deoxycortisol. These steroid changes exist in MACS on a spectrum, when patients are stratified according to the 1 mg-DST results, with higher post-dexamethasone serum levels of cortisol being associated with higher glucocorticoid secretion mirrored by lower androgen secretion[4]. Furthermore, in patients with MACS, higher cortisol levels after the 1 mg-DST were found to correlate with waist circumference and were also associated with several cardiovascular risk factors and drug-resistant hypertension prevalence in individuals with MACS[76,78]. Individuals with NFAT also show steroid metabolism changes similar to MACS, which supports the hypothesis that a subset of NFATs is associated with subtle autonomous cortisol secretion that results in reduced adrenal androgen output.

One study analysed the intratumoural steroid metabolome of individuals with MACS[85] (Supplementary Fig. 2c). Surprisingly, tissue concentrations of core steroid precursors (that is, corticosterone and 18OH-corticosterone) were higher in patients with MACS compared with those with adrenal Cushing syndrome; however, the reasons for these differences in intratumoural steroid flux are unclear. Other studies found stepwise increased expression (with lowest expression in NFAT, increased expression in MACS and highest expression in Cushing syndrome) of the cortisol-producing enzyme CYP11B1 in tumour tissue from patients with MACS or adrenal Cushing syndrome[86–88]. Moreover, in these patient populations, an inverse correlation exists between the degree of autonomous cortisol secretion and the intratumoural generation of classic androgens, in line with the lack of adrenal ACTH stimulation[86].

Targeted blood metabolomics

Three studies have assessed metabolic changes in the plasma and serum of patients with NFAT, MACS or Cushing syndrome[89–91] (Supplementary Fig. 3). Increasing autonomous cortisol secretion correlates with increasing perturbations of glycerophospholipids, specifically phosphatidylcholines and phosphatidylethanolamines[89–91]. Glycerophospholipids are the most abundant structural components of cell membranes and are involved in the regulation of lipid metabolism, lipoprotein secretion and whole-body energy metabolism, by supporting mitochondrial function[92]. Abnormal levels of glycerophospholipids in mitochondrial membranes have been linked to the development of mitochondrial dysfunction, abnormal glucose disposal, cardiovascular disease and the metabolic syndrome[93–103].

Increasing autonomous cortisol secretion was also associated with the dysregulation of several bioactive lipid classes including lysoglycerophospholipids, ceramides and sphingolipids[89–91]. These bioactive lipids regulate the cell cycle, immunomodulation, angiogenesis, intracellular trafficking, insulin signalling and endothelial function[104,105]. Of note, abnormal lipid-dependent cell signalling has been associated with multiple chronic inflammatory diseases including obesity, type 2 diabetes mellitus (T2DM), non-alcoholic fatty liver disease, cardiovascular events and cancer[103,105–111]. Interestingly, intratumoural metabolomics of MACS and adrenal Cushing syndrome found abnormal levels of several lysoglycerophospholipids and their precursors (glycerophospholipids), which correlated not only with CYP11B1 expression but also with tumour size[86]. These findings suggest a potential role of these lipid metabolites in the pathophysiology of autonomous cortisol secretion.

Alterations in circulating levels of acylcarnitines (intermediate products of β-oxidation) were observed in patients with MACS or Cushing syndrome[89–91]. This change points towards dysfunctional lipid β-oxidation and mitochondrial stress, which are associated with the risk of developing insulin resistance and T2DM[103,112,113].

Pathway enrichment analysis identified abnormal blood concentrations of several amino acids in individuals with MACS or Cushing syndrome[89–91] (Supplementary Fig. 3), including changes in aromatic amino acids (tyrosine and tryptophan), branched-chain amino acids (valine and isoleucine) and derivatives of citric acid cycle intermediates (aspartate, glutamate and asparagine), as well as arginine, proline, histidine, threonine and taurine. Alterations of these pathways have been linked to cardiovascular disease, the development of the metabolic syndrome, pro-inflammatory changes, sarcopenia, mitochondrial dysfunction and oxidative stress[114–122]. Arginine is a precursor of polyamines, involved in oxidative stress and cortisol-related immunomodulation[123]. Of note, increased circulating levels of the polyamine spermidine have been observed in individuals with Cushing syndrome[89,90] and linked to the presence of catabolic signs of cortisol excess such as proximal myopathy, skin thinning and easy bruising[89]. Similarly, blood concentrations of histidine have been found to negatively correlate with the severity of autonomous cortisol secretion and with catabolic signs including proximal myopathy[89].

Diagnosis of MACS

Patients with newly diagnosed adrenal masses should undergo a systematic evaluation to rule out malignancy and hormone excess. We refer the reader to current guidelines for the imaging and hormonal evaluation of adrenal incidentalomas[1]; in this section, we focus only on MACS.

Laboratory tests

Several diagnostic criteria for MACS have been proposed over the years, which include the non-suppressibility of serum cortisol after dexamethasone administration; either alone or in combination with other hormonal abnormalities, including elevated midnight serum cortisol, 24-h urinary free cortisol and low serum ACTH or dehydroepiandrosterone sulfate[3,124,125]. However, head-to-head comparisons between different diagnostic criteria are lacking and in 2016, the European Society of Endocrinology (ESE)–European Network for the Study of Adrenal Tumors (ENSAT) guidelines on adrenal incidentalomas recommended the use of

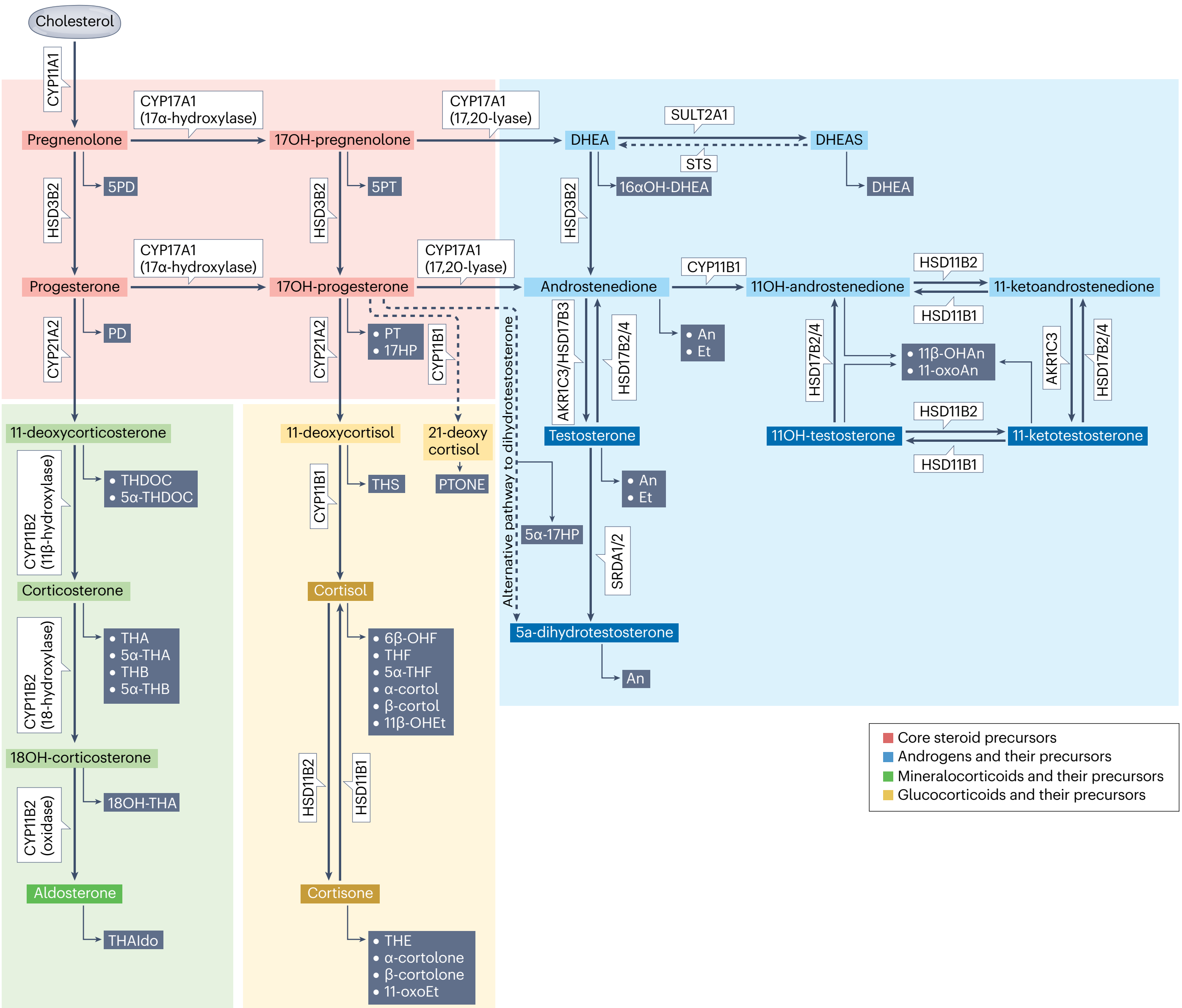

Fig. 3 | Schematic overview of steroidogenesis and corresponding urine steroid metabolites. Core steroid precursors and the pathways leading to the generation of mineralocorticoids, glucocorticoids and androgens are colour-coded. Arrows are labelled with catalysing enzymes and isoforms where appropriate. The urine metabolites of the various steroids are shown in grey. Both endogenous and exogenous steroids undergo phase I reactions primarily in the liver, which alter their biological activity by adding or revealing functional groups that can function as targets for subsequent conjugation (phase II) reactions. Ultimately, this results in steroid inactivation and increases water solubility, facilitating urinary excretion, which accounts for ~80% of steroid metabolite excretion. AKR, aldo-keto reductase; CYP, cytochrome P450; HSD, hydroxysteroid dehydrogenase; STS, steroid sulfatase; SULT, sulfotransferase. Steroid metabolite abbreviations: An, androsterone; DHEA, dehydroepiandrosterone; DHEAS, dehydroepiandrosterone sulfate; Et, etiocholanolone; PD, pregnanediol; PT, pregnanetriol; PTONE, pregnanetriolone; THA, tetrahydro-11-dehydrocorticosterone; THAldo, 3α5β-tetrahydroaldosterone; THB, tetrahydrocorticosterone; THDOC, tetrahydro-11-deoxycorticosterone; THE, tetrahydrocortisone; THF, tetrahydrocortisol; THS, tetrahydro-11-deoxycortisol; 5PD, pregnenediol; 5PT, pregnenetriol; 17HP, 17α-hydroxypregnanolone; 5α-17HP, 17α-hydroxy-3α,5α-pregnanolone; 6β-OHF, 6β-hydroxycortisol; 5α-THF, 5α-tetrahydrocortisol; 11β-OHEt, 11β-hydroxyetiocholanolone; 11-oxoEt, 11-oxoetiocholanolone; 16α-DHEA, 16a-hydroxy-DHEA; 11β-OHAn, 11β-hydroxyandrosterone; 11-oxoAn, 11β-oxoandrosterone; 5α-THDOC, 5α-tetrahydro-11-deoxycorticosterone; 5α-THA, 5α-tetrahydro-11-dehydrocorticosterone; 5α-THB, 5α-tetrahydrocorticosterone.

only one test to diagnose MACS: the 1 mg-DST[126]. This recommendation was confirmed by the recently published 2023 ESE–ENSAT guidelines[1].

MACS is defined by the failure to suppress serum cortisol sufficiently after overnight administration of 1 mg of dexamethasone in the 1 mg-DST (cortisol >50 nmol/l (>1.8 μg/dl)) in an individual with one or more adrenocortical tumours and who lacks the clinical signs of Cushing syndrome. Owing to the frequency of MACS and its potential clinical consequences, a 1 mg-DST is recommended in all individuals with adrenal incidentalomas. However, in frail adults with limited life expectancy, this test might not be warranted, as a targeted treatment for MACS might not be appropriate in this scenario[1].

The result of the 1 mg-DST should be interpreted as a continuum, as higher levels of cortisol associated with a higher risk of morbidity and mortality in large-scale studies[4,5,127] (Fig. 2); however, establishing this risk at the individual level is difficult. Other biochemical tests (for example, 24-h urinary free cortisol or late-night salivary cortisol) could be considered to assess the degree and severity of the cortisol excess, although the results are within the normal range in most patients with MACS and the 2023 ESE–ENSAT guidelines do not recommend their routine use[1].

The 1 mg-DST should not be repeated routinely unless a false-positive result is suspected, as the possibility of 'true' MACS resolving over time in a patient is deemed to be very low[3]. Similarly, patients with NFAT should undergo repeat hormonal testing only if they develop or have a worsening of potential cortisol-related comorbidities during follow-up[1].

Several factors can affect the reliability of the 1 mg-DST. First, the clinician should ensure that the individual under investigation carried out the test correctly and took dexamethasone as directed. The measurement of serum levels of dexamethasone can improve the accuracy of the 1 mg-DST[125,128], but this test is not currently available in most centres. Furthermore, the 50 nmol/l (1.8 μg/dl) cut-off was established using older radioimmunoassays; using different, more modern serum cortisol immunoassays during the 1 mg-DST caused both false-positive and false-negative results compared with the reference standard liquid chromatography–tandem mass spectrometry[129]. False-positive results are also possible in individuals with elevated serum levels of cortisol-binding globulin (caused by oral oestrogens, selective oestrogen receptor modulators, pregnancy or chronic active hepatitis), active psychiatric disorders, uncontrolled diabetes mellitus or alcohol abuse or taking strong CYP3A4 inducers (which decrease the bioavailability of dexamethasone)[8,125]. In these situations, salivary cortisone[130] or the synchronous measurement of free cortisol during the 1 mg-DST[125] could aid the interpretation of the test. False-negative 1 mg-DST results are less of a concern, but should be considered during treatment with strong CYP3A4 inhibitors (which increase the bioavailability of dexamethasone) or in conditions potentially linked with low blood levels of cortisol-binding globulin (for example, proteinuria)[8].

Clinical presentation

Although the absence of overt features of Cushing syndrome is in the definition of MACS, in practice, the clinical presentation and physical examination could have overlapping features that make it challenging to confidently diagnose Cushing syndrome versus MACS. For example, when an experienced clinician is assessing the presence of individual physical examination features using a standardized checklist, abdominal obesity, supraclavicular or dorsocervical pads, rounding of the face and skin changes were recorded in 78–90% of patients with Cushing syndrome, but also in 21–38% of patients with MACS[18]. The clinical severity score (a combination of physical exam features and cortisol-associated comorbidities) was lower in patients with MACS versus Cushing syndrome (7 versus 15)[18], potentially providing a more accurate distinction between the two conditions. Although MACS almost never progresses towards Cushing syndrome[3], long-standing milder hypercortisolism could potentially lead to the development of physical examination features that are more difficult to diagnose and patients are less likely to report these changes. For example, when assessed with a sit-to-stand test and hand grip test, and after age adjustment and sex adjustment, patients with MACS demonstrated similar impairment in muscle strength as patients with Cushing syndrome; however, self-reported muscle weakness was higher in Cushing syndrome (67% versus 48% in MACS)[18].

Comorbidities and complications of MACS

Cardiometabolic health

In two population-based studies, patients with adrenal incidentalomas presented with a higher prevalence and incidence of cardiovascular risk factors including hypertension, dysglycaemia, peripheral vascular disease, chronic kidney disease, cardiac arrhythmias, heart failure, cardiovascular and thromboembolic events when compared with control individuals[131–133].

Compared with patients with NFAT, those with MACS had a higher prevalence and incidence of cardiometabolic risk factors, with the data published over the past decade systematically summarized in the recent guidelines and the accompanying meta-analysis[1,3–6,25,133–136]. Patients with MACS were reported to have a higher prevalence of T2DM, hypertension and dyslipidaemia than patients with NFAT[134]. In a cross-sectional study of prospectively and consecutively recruited patients with MACS, >50% of patients with MACS had obesity, with a median BMI of 31 kg/m^2 (ref. 18). Patients with MACS were more likely to develop new cardiovascular events than patients with NFAT during conservative follow-up[3,5,137,138]. Notably, even patients with NFAT demonstrate a higher cardiometabolic risk than individuals without adrenal adenomas[7,22], especially those with 1 mg-DST cortisol closer to the MACS threshold of 50 nmol/l (1.8 μg/dl)[4,25,139].

Mortality

In a population study of 1,004 patients with adrenal incidentalomas from Olmsted County, MN, USA, mortality was not different in patients versus control individuals[132]. Another, much larger study of 17,726 patients with adrenal adenomas without hormone excess from Sweden found a higher overall mortality in patients with adrenal incidentaloma (HR 1.21, 95% CI 1.16–1.26), in both men and women, and especially in younger patients <65 years age, with mortality mainly owing to cardiovascular and cancer causes[131]. Several studies examined mortality based on the 1 mg-DST cortisol level. In a retrospective, matched cohort study of 1,154 patients with adrenal incidentaloma from Sweden, no difference in mortality was noted between patients with NFAT and control individuals, whereas patients with MACS had higher mortality, especially in younger patients and patients with 1 mg-DST cortisol >80 nmol/l (3.0 μg/dl)[140]. In another retrospective multicentre study of patients with NFAT and MACS, all-cause mortality adjusted for age, sex, comorbidities and previous cardiovascular events was increased in patients with MACS and correlated with 1 mg-DST cortisol levels (HR 1.52 (95% CI 1.19–1.94) for 1 mg-DST cortisol 50–138 nmol/l (1.8–5.0 μg/dl); HR 1.77 (95% CI 1.20–2.62) for 1 mg-DST cortisol >138 nmol/l (>5.0 μg/dl)). Notably, the increase in mortality was only observed in women[5]. Excessive mortality was mainly owing to cardiovascular and cancer causes, but also owing to infection, suggesting an effect of MACS on immunity[5].

In a meta-analysis published in 2023 that included 5,921 patients, all-cause mortality adjusted for confounders was increased in patients with MACS versus patients with NFAT (HR 1.54 (95% CI 1.27–1.81))[134].

Bone health

In a population-based study of patients with adrenal adenomas without overt hormone excess, the prevalence of vertebral fractures was 6.4% versus 3.6% in control individuals and the overall prevalence of fragility fractures was 16.6% versus 13.3%[141]. Patients with adrenal adenomas were also at a higher risk of developing a new fracture than individuals without adenomas during a follow-up of 7 years (HR 1.3; 95% CI 1.1–1.5), after adjusting for common confounding factors[141]. Notably, as a fracture was defined based on the diagnostic code, both prevalence and incidence estimates reflect clinically important fractures. A meta-analysis published in 2023 reported 10 studies investigating the prevalence of vertebral fractures in MACS versus NFAT; however, pooled analysis was not possible owing to heterogeneity regarding outcomes and cut-offs for MACS[134]. In a more recent cross-sectional study that was not included in the meta-analysis, the prevalence of vertebral fractures (most of these were asymptomatic) was higher in MACS (34% versus 24% in NFAT). This difference was statistically significant in post-menopausal women, but not in men. In post-menopausal women, the prevalence of fragility fractures was also associated with age, smoking and 1 mg-DST cortisol levels[142].

Assessing the risk of fracture in patients with MACS is challenging, and the standard assessments of bone mineral density with dual-energy X-ray absorptiometry do not accurately identify patients at risk, probably owing to MACS having a higher effect on bone quality than density. Trabecular bone score is a grey-level texture measurement that can be applied to dual-energy X-ray absorptiometry images and was previously reported to correlate with parameters of bone microarchitecture, including trabecular number, connectivity density and trabecular spacing[143]. In patients with adrenal adenomas, trabecular bone score correlated with the 1 mg-DST cortisol measurement and the ratio of serum cortisol to dehydroepiandrosterone sulfate and was associated with both prevalent and incident fractures[144–146].

Frailty and muscle strength impairment

Patients with MACS demonstrate hormonal abnormalities that mimic those seen with normal ageing, with a decrease in androgens, a slight increase in glucocorticoid production and a blunted circadian cortisol rhythm. Moreover, patients with MACS demonstrate a higher prevalence of age-associated morbidities[3,6]. Hypothesizing that MACS is a state of accelerated ageing, Singh et al.[135] developed a frailty index assessment for patients with adrenal adenomas that is based on a 47-variable deficit model, with a range between 0 and 1 and a cut-off of >0.25 being consistent with frailty. The frailty index was applied to 168 patients with MACS and 275 patients with NFAT and demonstrated a higher prevalence of frailty in MACS (24% versus 18% in NFAT). In a subsequent prospective multicentre study using the same frailty index, the prevalence of frailty in patients with adrenal adenomas was increased above a 1 mg-DST cortisol threshold level of 28 nmol/l (1.0 μg/dl)[6]. The frailty index was associated with assessments of muscle strength, such as the 6-min walk test, gait speed, sit-to-stand test and hand grip measurements[6]. Another study examined hand grip and sit-to-stand tests in patients with MACS and Cushing syndrome. Notably, standardized *Z* scores for both tests were similar in MACS and Cushing syndrome, suggesting comparable impairments in muscle strength[18]. Patients with MACS also had lower hand grip scores when compared with age-matched and sex-matched control individuals[18].

Quality of life, sleep, mood and cognition

Limited data suggest impaired quality of life, sleep disturbances and a high prevalence of psychiatric disorders in patients with adrenal adenomas overall and more so in those with MACS versus NFAT[6,135,147–149]. For example, patients with MACS were found to have a higher score on the Beck Depression Inventory (which measures symptoms of depression) and a lower mental and physical component summary score on the Short Form-36 survey (a quality-of-life assessment) compared with control individuals[148]. In a larger study, patients with MACS reported a lower quality of life than control individuals when assessed with the Short Form-36 survey for both physical and mental component summary scores, as well as all subdomains[18]. In addition, patients with MACS have a higher prevalence of insomnia and sleep difficulties compared with patients with NFAT[135,149]. Several studies reported no differences in the prevalence of psychiatric disorders in MACS and NFAT[135,149]; however, a larger cross-sectional study showed a higher prevalence of self-reported depression and anxiety in patients with MACS compared with control individuals[6]. Furthermore, in a population-based study, the incidence of depression, anxiety and sleep disorders was increased in patients with adrenal incidentalomas compared with control individuals[150].

Data on the effect of MACS on cognition are discrepant. One small study showed better cognitive function in verbal fluency and executive function in patients with MACS versus NFAT[149]. By contrast, another study found that patients with MACS demonstrated worse performance in working memory and visuospatial–constructional domain than those with NFAT[147].

Management of MACS

Adrenalectomy versus conservative follow-up for control of comorbidities

Adrenal incidentaloma guidelines from 2023 recommend considering adrenalectomy after discussion by an expert multidisciplinary team in patients with unilateral MACS and comorbidities potentially related to MACS, after considering individual factors such as age, tumour characteristics, availability of surgical expertise and patient preference[1] (Fig. 4).

A meta-analysis published in 2016, of mostly observational studies, reported improvements in hypertension and T2DM but not dyslipidaemia in patients with MACS treated with adrenalectomy compared with conservative management[151]. However, a subsequent observational study reported post-adrenalectomy improvements in hypertension and dyslipidaemia, but not in T2DM[137]. A 2022 randomized study of 62 patients with MACS showed that adrenalectomy led to more improvements in hypertension (68% versus 13%) and glycometabolic control (28% versus 3%) compared with conservative management[152]. Furthermore, another meta-analysis of patients with MACS and NFAT from 2023 reported an improvement in hypertension, glycometabolic control and dyslipidaemia following adrenalectomy[134]. Overall, the evidence suggests that adrenalectomy is effective for controlling the cardiometabolic complications of unilateral MACS. Yet, it is important to note that studies investigating the effectiveness of adrenalectomy in MACS have multiple limitations, including being mostly of retrospective design, using heterogeneous definitions of MACS, variable timelines of assessment and definitions of studied outcomes and a lack of optimal comparative groups.

Challenges in recommending adrenalectomy

Considering the high prevalence of MACS, uncertain causality between MACS and comorbidities, as well as difficulties in predicting the degree of post-surgical improvement in comorbidities[4,134], identifying patients who are most likely to benefit from adrenalectomy is challenging. The duration of untreated MACS before the diagnosis could affect the reversibility of comorbidities with adrenalectomy; however, as most patients with MACS are discovered incidentally[2,4], the duration of MACS is impossible to estimate. Most studies concentrated on the reversibility of hypertension and T2DM[134], and the effectiveness of adrenalectomy on other clinical outcomes, including quality of life, was scarcely explored. Finally, treatment of MACS before development of comorbidities might be more effective in mitigating morbidity and mortality risk, but this approach has no evidence to date; especially as the guidelines[1] recommend adrenalectomy only in patients with MACS who have established cardiovascular comorbidities.

Postoperative adrenal insufficiency, glucocorticoid withdrawal syndrome and medical management

The incidence of post-operative adrenal insufficiency after unilateral adrenalectomy for MACS is around 50%, with a mean time to recovery of 6.5 months[153]. All patients should be counselled before surgery about the risk of adrenal insufficiency and its management. Two approaches can be used in the hospital setting to manage potential post-operative adrenal insufficiency. First, uniform perioperative glucocorticoid therapy can be administered with outpatient reassessment of adrenal function, as recommended in the 2023 guidelines for adrenal incidentalomas[1]. Second, adrenal function can be tested before discharge from the hospital and glucocorticoid therapy initiated only in patients with documented adrenal insufficiency[154,155].

Patients should also be counselled about the glucocorticoid withdrawal syndrome that can occur after adrenalectomy; symptoms include arthralgias, myalgias, fatigue, decreased appetite and mood changes[155,156]. The pre-adrenalectomy clinical severity score was reported to be associated with the severity of glucocorticoid withdrawal syndrome[156] and could be helpful in counselling patients and individualizing management. In patients with post-operative adrenal insufficiency, symptoms of glucocorticoid withdrawal were reported to subside once the morning serum concentration of cortisol before taking oral glucocorticoid is ≥276 nmol/l (≥10 μg/dl)[155].

Following adrenalectomy, patients with MACS might show improvements in hypertension and T2DM. Patients should be advised on blood pressure and glucose monitoring, as the number and the dose of antihypertensive and/or diabetes mellitus medications might need to be decreased to avoid hypotension and hypoglycaemia[151,157].

Special situations: bilateral adrenal masses

Around 12–30% of patients with MACS have bilateral adrenal masses[3–5]. Patients with bilateral MACS present with a higher cardiometabolic burden than patients with unilateral MACS, which could potentially be explained by the differences in the degree of MACS[4]. The imaging phenotype can be consistent with bilateral adrenal adenomas or PBMAH (Fig. 5). In a patient with bilateral adrenal adenomas, only one of the adenomas could be cortisol-secreting, whereas patients with PBMAH always have bilateral MACS. Careful review of cross-sectional imaging is frequently helpful to distinguish between the two aetiologies.

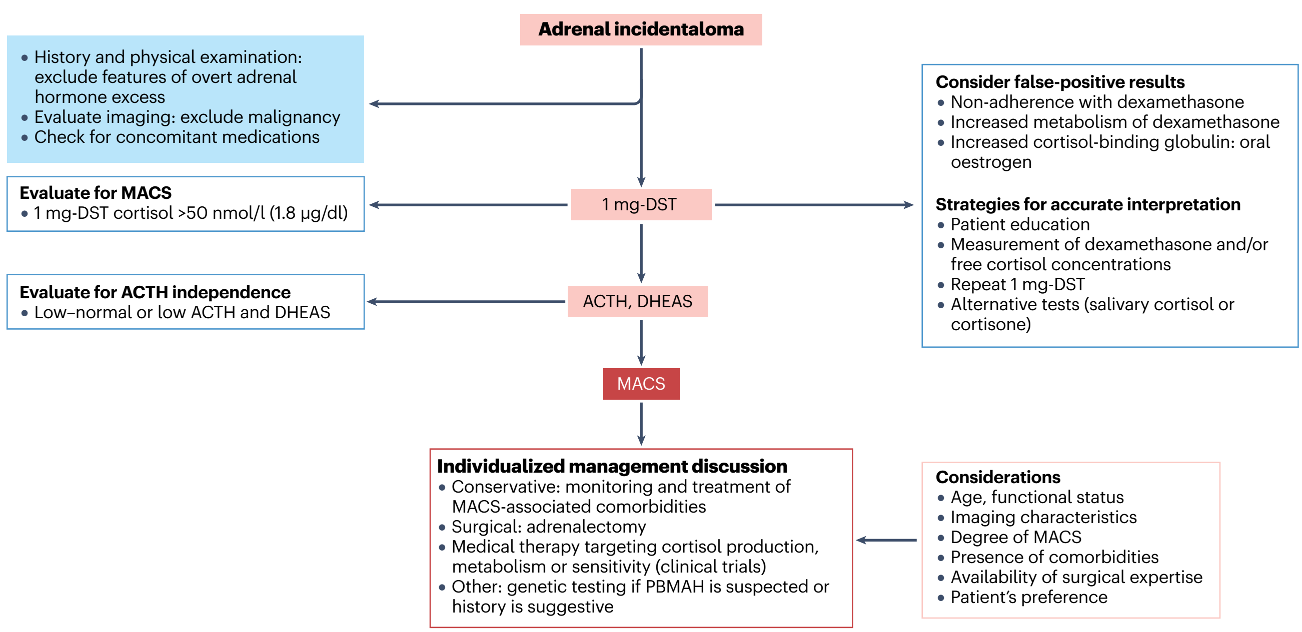

Fig. 4 | Diagnosis and management of mild autonomous cortisol secretion. Diagnosis of mild autonomous cortisol secretion (MACS) is based on an abnormal 1 mg overnight dexamethasone suppression test (1 mg-DST) measurement of serum levels of cortisol >50 nmol/l (>1.8 μg/dl) in a patient with unilateral or bilateral adrenal nodules and in the absence of the features of overt Cushing syndrome[1]. False-positive results for 1 mg-DST should be considered (for example, when suspecting noncompliance with the test protocol, decreased dexamethasone absorption or increased dexamethasone metabolism). The 2023 guidelines on adrenal incidentalomas recommend repeating 1 mg-DST before intervention (such as surgery)[1]. If 1 mg-DST cortisol is abnormal, evaluating for ACTH independence is the next step, by measuring blood levels of adrenocorticotropic hormone (ACTH) and dehydroepiandrosterone sulfate (DHEAS). DHEAS levels are generally lower in women than in men and physiologically decline with age; therefore, DHEAS should be interpreted cautiously in post-menopausal women with MACS, who comprise the majority of cases. Other tests for cortisol excess, such as 24-h urine cortisol or midnight salivary cortisol, are less helpful for many patients with MACS, who demonstrate results within normal ranges. Management of MACS is individualized after considering age, the functional status of the patient, the degree of MACS, comorbidities potentially attributable to MACS, availability of surgical expertise and patient preference. PBMAH, primary bilateral macronodular adrenal hyperplasia.

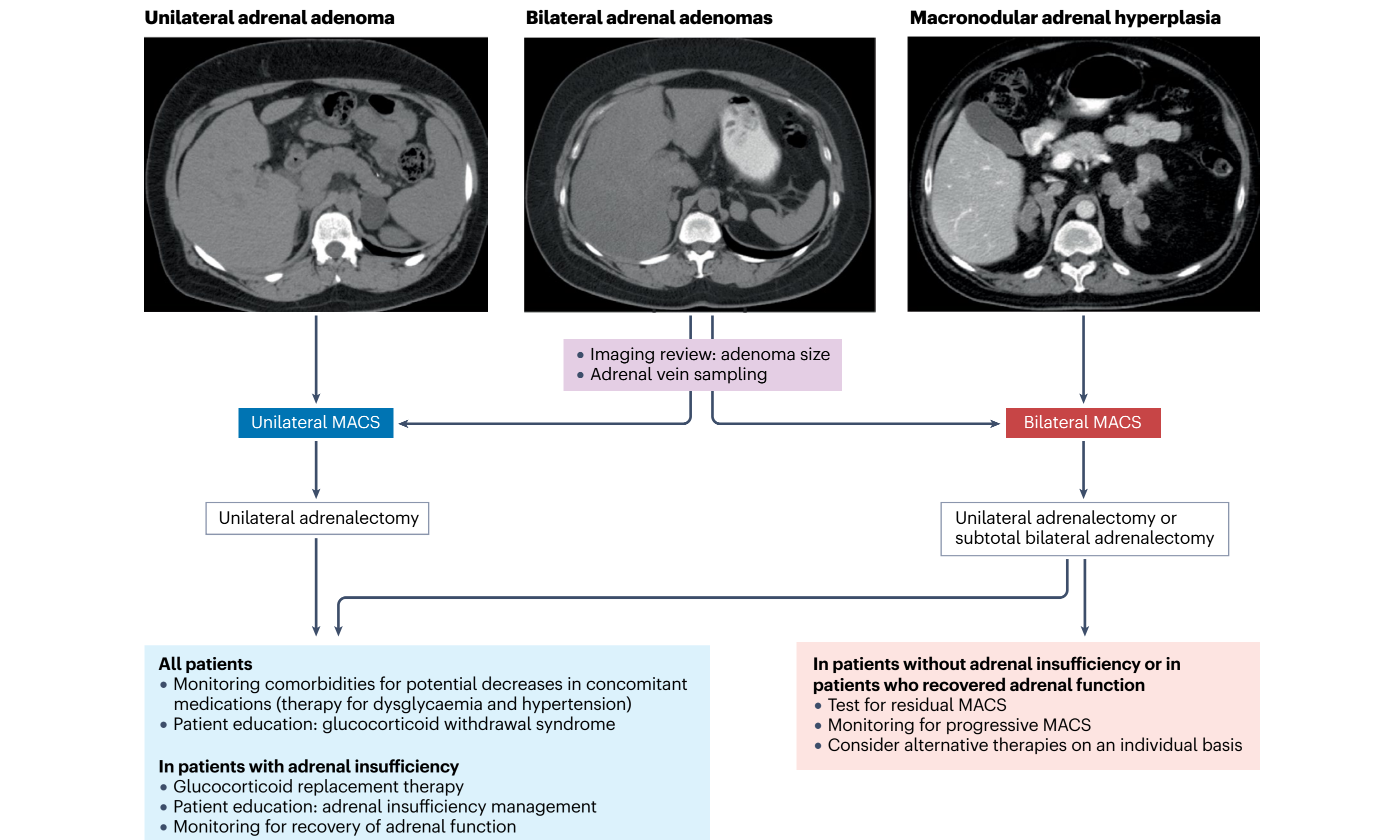

Fig. 5 | Surgical management in patients with mild autonomous cortisol secretion. The approach to surgical management in patients with mild autonomous cortisol secretion (MACS) differs between patients with MACS owing to unilateral adenoma (which is more common) and patients with bilateral adrenal nodules. In patients with unilateral MACS, unilateral adrenalectomy could lead to post-operative adrenal insufficiency. In these patients, glucocorticoid replacement therapy should be initiated and education on sick day rules performed. Monitoring for adrenal function should be organized, with most patients recovering within 1-year post-adrenalectomy[155]. In patients with bilateral adrenal adenomas, adrenal vein sampling could identify whether MACS is unilateral or bilateral, although this procedure is not uniformly available and the interpretation varies. If unilateral MACS is diagnosed in patients with bilateral adenomas, adrenalectomy will be curative, and management is similar to patients with unilateral MACS owing to unilateral adenoma. In patients with bilateral adenomas or primary bilateral macronodular adrenal hyperplasia (PBMAH), unilateral adrenalectomy might induce a temporary remission of MACS (and even adrenal insufficiency).

Patients with PBMAH are more likely than those with adenomas to demonstrate a combined adrenal hormone excess, mainly with MACS and primary aldosteronism, and rarely with sex steroid excess[17]. Guidelines suggest individualization of management options in patients with bilateral adenomas or PBMAH based on age, sex, degree of MACS, general condition, comorbidities and patient preference[1]. Several studies investigated the efficacy of unilateral adrenalectomy in patients with PBMAH and Cushing syndrome but not with PBMAH and MACS[158–160]. These studies demonstrate that unilateral adrenalectomy might induce remission or improve the degree of hypercortisolism in patients with PBMAH or bilateral cortisol-secreting adenomas[158–160]. Notably, unilateral adrenalectomy in a patient with bilateral MACS might not restore normal circadian cortisol secretion even if the post-adrenalectomy 1 mg-DST cortisol concentration is <50 nmol/l (<1.8 µg/dl), owing to the remaining subtle glucocorticoid autonomy.

In a patient with MACS with a clear imaging phenotype of PBMAH, unilateral adrenalectomy usually targets the larger adrenal gland (if asymmetric disease) or the adrenal gland that is technically easier to remove (if symmetric bilateral disease). Adrenal insufficiency can occur depending on the degree of preoperative MACS and the amount of adrenal tissue removed and, as such, should be anticipated and planned for (Fig. 5). In patients with PBMAH treated with unilateral adrenalectomy, subsequent management includes monitoring for recurrence of MACS and associated comorbidities[1].

In a patient with bilateral adenomas, adrenal vein sampling might be used to identify the cortisol-secreting lesion. However, only a handful of studies on the use of adrenal vein sampling in patients with MACS have been published to date[161,162]. Reported adrenal vein sampling procedures varied in the dexamethasone dose administered before testing and how a successful canulation was defined[161,162]. In a 2023 study of 25 patients with MACS with bilateral lesions undergoing adrenal vein sampling for cortisol excess, 20 patients were treated with unilateral adrenalectomy and experienced no residual or recurrent MACS during follow-up[162]. In these patients, post hoc analysis showed that an adrenal vein to inferior vena cava cortisol ratio of >9 and cortisol lateralization ratio of >2.3 had a higher specificity for unilateral MACS[162].

Future directions

One of the challenges in the evaluation of patients with MACS is the unclear causality between MACS and concomitant comorbidities. As the prevalence of cardiovascular risk factors, such as obesity, hypertension and dysglycaemia, is common in the general population, the contribution of MACS to the development of these comorbidities is difficult to assess. As MACS is discovered incidentally, the duration of MACS is unknown at the time of diagnosis, further making determination of the clinical history and timeline for the development of comorbidities difficult. Identifying biomarkers indicative of the individual clinical outcomes of MACS will not only aid in pinpointing patients who are likely to improve MACS-related comorbidities post-adrenalectomy but could also identify patients who could be treated preventively, before developing comorbidities.

Another gap is the lack of large randomized clinical trials that test the efficacy of adrenalectomy on patient-important outcomes such as quality of life, sleep, mood, fractures and cognition, in addition to cardiometabolic health and mortality.

Therapy targeting cortisol secretion, metabolism or sensitivity might be effective in MACS; however, studies demonstrating the efficacy and safety of long-term medical therapy in patients with MACS are lacking. Only several small case series were published to date using metyrapone, mifepristone or 11β-hydroxysteroid dehydrogenase type 1 inhibitors in patients with MACS; several studies showed improvements in glucose metabolism[163,164] and another study using metyrapone showed a reduction in the cardiovascular risk marker IL-6 (ref. 165). Short-term medical therapy might have a role in identifying patients who are likely to benefit from adrenalectomy. By contrast, long-term medical therapy, assuming a favourable safety profile, might be preferred in patients with bilateral MACS or those who are not surgical candidates. Considering that the number of patients diagnosed with adrenal incidentalomas and MACS continues to increase over time, the potential role of medical therapy is likely to broaden in the future. Prospective clinical trials are needed to establish the feasibility of medical therapy for the long-term management of MACS.

Conclusions

MACS is prevalent among patients with adrenal incidentalomas, with patients demonstrating multiple associated comorbidities and increased mortality. Management in patients with MACS should be individualized. Although unilateral adrenalectomy is curative in patients with MACS owing to unilateral adenoma, management of bilateral MACS is more challenging, frequently requiring a multimodal approach and lifelong monitoring. All patients with MACS treated with adrenalectomy need a post-operative plan that includes management of possible adrenal insufficiency.

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